Stars initially form from clouds of dust and gas…….

The force of gravity makes the dust particles come spiraling in together. As they do, gravitational energy is converted into heat energy and the temperature rises…….

When the temperature gets high enough, hydrogen nuclei undergo nuclear fusion to form helium nuclei and give out massive amounts of heat and light. A star is born…….

The star immediately enters a long stable period where the heat created by the nuclear fusion provides an outward pressure to balance the force of gravity pulling everything inwards. In this stable period it's called a MAIN SEQUENCE STAR……..

Eventually the hydrogen begins to run out and the star then swells into a RED GIANT. It becomes red because the surface cools…….

A small star like our Sun will then begin to cool and contract into a WHITE DWARF and then finally as the light fades completely it becomes a BLACK DWARF……..

Big stars start to glow brightly again as they undergo more fusion and expand and contract several times forming heavier elements in various nuclear reactions. Eventually they explode in a SUPERNOVA…….

The exploding supernova throws the outer layers of dust and gas into space leaving a very dense core called a NEUTRON STAR. If the star is big enough this will become a BLACK HOLE……...

The dust and gas thrown off by the supernova will form into SECOND GENERATION STARS like our Sun……..